ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** parallel to the incident ray but opposite in direction	For 90°, reversed.
2	**(b)** umbra	Complete shadow.
3	**(b)** virtual image position follows the law of reflection	Geometric consequence.
4	**(a)** 20°	Twice the turn of the mirror (10 x 2).
5	**(b)** shadow	Dark.
6	**(c)** red	Same colour.
7	**(d)** 0.8 m	Half.
8	**(d)** double	Distance ratio 2:1.
9	**(c)** black	Absence of light.
10	**(a)** 12 m	2 x 6.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle of incidence is 0° (1); The ray retraces its path (1).	1+1
12	The Sun is low in the evening (1); Rays fall at a smaller angle so the shadow stretches (1).	1+1
13	It reflects sunlight (1); Surface scatters light (1).	1+1
14	5 cm (1); Virtual, erect and laterally inverted (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	i = r = 55° (1); Angle between rays 110° (1); Angle with mirror 35° (1).	1+1+1
16	Fix a stick vertically (1); Mark shadow lengths (1); Shortest shadow is at noon (1).	1+1+1
17	Slightly curved (1); Less polished (1); Distorted image (1).	1+1+1
	OR: Initially 8 m (1); Now 3 m, so 6 m (1); Decrease 2 m (1).	1+1+1
18	Reflect headlights (1); Improve visibility (1); Safety (1).	1+1+1
	OR: No light (1); Nothing to reflect (1); Need a source (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Number = 360/60 - 1 = 5 (1); Each mirror forms images (1); Images reflected again (1); Place object and count (1); Vary angle (1).	1+1+1+1+1
	OR: Pinhole real inverted (1); Mirror virtual erect (1); Pinhole needs a screen (1); Mirror image behind (1); Both use straight-line light (1).	1+1+1+1+1
20	Place object at several distances (1); Find image position by parallax with pin (1); Record distances (1); Compare in a table (1); Equal (1).	1+1+1+1+1
	OR: First mirror shows front (1); Second mirror behind (1); Reflect image of back (1); Repeated images (1); Customer sees the back (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 50 cm	20 x 100/40.
2	**(c)** black	Absorbs red.
3	**(c)** 2 m behind the mirror	Equal distance.
4	**(a)** 50 m	40/0.8.
5	**(b)** red	Reflects red.
6	**(d)** 50°	Half.
7	**(d)** 9 m	12 x 1.5/2.
8	**(a)** paper reflects light diffusely	Diffused reflection.
9	**(c)** 3	360/90 - 1.
10	**(a)** 20 cm	10 x 1/0.5.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Image of the Sun on a screen (1); Do not look at the Sun directly (1).	1+1
12	They give an erect image of the same size (1); Convex or concave would distort (1).	1+1
13	From 6 m to 10 m (1); Increases by 4 m (1).	1+1
14	Half her height (1); Independent of her distance from the mirror (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Height ratio = 15/30 (1); 0.5 x 6 (1); 3 cm (1).	1+1+1
16	Curved surface forms a different image (1); It is not a plane mirror (1); Image is inverted and smaller (1).	1+1+1
17	Write a word (1); Hold before a mirror (1); Letters reversed (1).	1+1+1
	OR: Ratio 1.2/0.9 (1); 30 x 1.333 (1); 40 m (1).	1+1+1
18	Place a screen behind the mirror (1); No image forms on it (1); Image seen only in the mirror (1).	1+1+1
	OR: Most light passes (1); A little is reflected and absorbed (1); Faint shadow (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Half of height: 0.8 m (1); Top edge halfway between eyes and top of head: 5 cm above eyes (1); Bottom edge halfway between eyes and feet (1); Mirror distance does not matter (1); Verify with diagram (1).	1+1+1+1+1
	OR: Umbrella reflects light (1); Reduces harsh shadows (1); Road reflectors (1); Cat's-eye (1); Helps visibility (1).	1+1+1+1+1
20	Ray enters top mirror (1); Reflects at 45° (1); Goes down to second mirror (1); Reflects to eye (1); Longer tube sees further (1).	1+1+1+1+1
	OR: Magnification = 90/30 = 3 (1); Diameter 15 cm (1); Area ratio 9:1 (1); Rays straight (1); Check (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** diffused reflection	Scatters.
2	**(b)** very large	Repeated reflection.
3	**(d)** 4 cm	200 x 20/1000.
4	**(d)** regular reflection	Water acts as a mirror.
5	**(c)** 1 cm	20 x 10/200.
6	**(d)** double	Inversely.
7	**(d)** a shadow is darker where no light reaches	Shadow is absence of light.
8	**(d)** laterally inverted	Reverse.
9	**(b)** sharp shadow	Straight.
10	**(b)** plane mirror	Reversal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rear-view mirror shows it correct (1); Easy to read (1).	1+1
12	A small source gives a sharp shadow (1); Extended source gives penumbra (1).	1+1
13	Glare dazzles drivers (1); Reduces visibility (1).	1+1
14	Change angle between mirrors (1); Count the images (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ratio 12/6 = 2 (1); Distance = 2 x 20 (1); 40 cm (1).	1+1+1
16	Cut windows (1); Fix mirrors at 45° (1); Look in the lower window (1).	1+1+1
17	Place a pin in front (1); Look from different sides (1); Line up the second pin (1).	1+1+1
	OR: Image is of the Sun, not the hole (1); Small hole (1); Circular Sun (1).	1+1+1
18	i changes by 20° (1); r also changes by 20° (1); Angle between rays changes by 40° (1).	1+1+1
	OR: Image larger (1); Dimmer (1); Same inversion (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sundial uses shadow of gnomon (1); Puppets use opaque cut-outs (1); Eclipses use Earth and Moon shadows (1); Rectilinear propagation (1); Different sizes (1).	1+1+1+1+1
	OR: Image moves at 1.5 m/s towards mirror (1); Relative 3 m/s (1); Distances equal (1); Both approach (1); Check with time (1).	1+1+1+1+1
20	Shadow tip is 5 + 2.4 = 7.4 m from the pole (1); Heights are in the ratio of distances: H/7.4 = 1.5/2.4 (1); H = 1.5 x 7.4/2.4 (1); About 4.6 m (1); Check the units (1).	1+1+1+1+1
	OR: Mirrors (1); Shiny utensils (1); Windows (1); Shadow on the wall (1); Conclusions (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.